

Node.js Course Outline

1. Node.js Fundamentals

- What is Node.js?
- Node.js vs JavaScript in the browser
- Installing Node.js
- Running `.js` files with Node
- `node` and `npm`
- `package.json`
- npm packages
- `node_modules`
- `package-lock.json`

2. Modules

- What are modules?
- Built-in modules
- `require()`
- `module.exports`
- Custom modules
- Importing/exporting your own files
- CommonJS vs ES Modules
- `import / export`

Example:

```
// math.js
const add = (a, b) => a + b;

module.exports = add;
// app.js
const add = require("./math");

console.log(add(5, 3));
```

3. File System (`fs`)

- What is `fs`?
- Reading files
- Writing files
- Appending files
- Deleting files
- Renaming files

- Creating directories
- Reading directories
- Synchronous vs asynchronous methods
- Callbacks with `fs`

Important methods:

`readFile()`
`writeFile()`
`appendFile()`
`unlink()`
`mkdir()`
`readdir()`
`rename()`

4. Paths (`path`)

- What is the `path` module?
 - File paths
 - Directory paths
 - `path.join()`
 - `path.basename()`
 - `path.dirname()`
 - `path.extname()`
 - `__dirname`
 - `__filename`
-

5. Events & Event-Driven Architecture

- Events in Node.js
 - `EventEmitter`
 - Creating custom events
 - Event listeners
 - `.on()`
 - `.emit()`
 - Why Node.js is event-driven
-

6. Asynchronous JavaScript in Node.js

You should be comfortable with:

- Callbacks

- Callback functions
- Callback hell
- Promises
- `.then()`
- `.catch()`
- `async/await`
- `try/catch`
- Error handling
- Event loop
- Non-blocking I/O

This is **very important** for Node.js.

7. HTTP Module

Before Express, understand Node's built-in HTTP server.

- `http` module
- Creating a server
- Request and response
- HTTP methods
- Status codes
- Headers
- Routes
- Query parameters
- Request body
- JSON responses

Example:

```
const http = require("http");

const server = http.createServer((req, res) => {
  res.write("Hello Node.js");
  res.end();
});

server.listen(3000);
```

8. npm & Packages

- Installing packages
- Local vs global packages
- Dependencies
- Dev dependencies

- `npm install`
 - `npm uninstall`
 - `npm update`
 - `npm scripts`
 - Semantic versioning
 - `package.json`
-

9. Express.js

Once the Node fundamentals are clear, move to Express.

- What is Express?
- Creating an Express server
- Routing
- Route parameters
- Query parameters
- Middleware
- Request/response objects
- Sending JSON
- HTTP status codes
- Router
- Serving static files
- Error-handling middleware

Example:

```
app.get("/users", (req, res) => {  
  res.json(users);  
});
```

10. REST APIs

- What is an API?
- REST architecture
- CRUD
- GET
- POST
- PUT
- PATCH
- DELETE
- JSON
- Status codes
- API testing with Postman

- Route structure
- Controllers

Example:

```
GET /users
GET /users/:id
POST /users
PUT /users/:id
DELETE /users/:id
```

11. Databases

Learn one database properly first.

MongoDB:

- MongoDB basics
- Collections
- Documents
- CRUD
- MongoDB Atlas
- Mongoose
- Schemas
- Models
- Validation
- Relationships
- `populate()`

Or, since you're already learning SQL:

MySQL/PostgreSQL:

- Connecting Node to database
 - Queries
 - CRUD
 - Parameterized queries
 - Relationships
 - Connection pooling
-

12. Authentication & Authorization

- Authentication vs authorization
- Password hashing
- `bcrypt`
- Sessions

- Cookies
 - JWT
 - Protected routes
 - Roles/permissions
 - Logout
 - Authentication middleware
-

13. Validation & Security

- Input validation
 - Error handling
 - Environment variables
 - `.env`
 - CORS
 - Helmet
 - Rate limiting
 - Password security
 - SQL/NoSQL injection
 - Secure API design
-

14. File Uploads

- Uploading files
 - `multipart/form-data`
 - Multer
 - Images
 - File size/type validation
 - Storing uploaded files
 - Cloud storage basics
-

15. Advanced Node.js

After you're comfortable building APIs:

- Event loop in depth
- Streams
- Buffers
- File streams
- Child processes
- Worker threads
- Clustering
- WebSockets

- Webhooks
 - Caching
 - Redis
 - Queues/background jobs
-

16. Testing & Deployment

- Unit testing
- Integration testing
- API testing
- Jest/Vitest
- Environment variables
- Production configuration
- Logging
- Deployment
- Docker basics
- CI/CD basics